Cross-Linguistic Transfer Diagnostics for Low-Resource Financial NLP in Indic Languages

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***Abstract*—**Multilingual language models transfer well on general language-understanding benchmarks, but it is not known whether they preserve financial meaning across Indic languages. IndicFinNLP was built for single-language baselines, and its language splits differ in topic, entity and difficulty, which confounds any comparison across languages. We build a controlled parallel corpus from IndicFinNLP in Hindi, Bengali, Telugu and Malayalam, translating each native split directly between Indic languages with IndicTrans2 into 92,760 machine-translated rows with labels unchanged. Every row is checked for numeral preservation, structural integrity and LaBSE similarity against a threshold calibrated on human translations, and is flagged rather than dropped. An independence audit shows that the sustainability and ESG-theme splits are already parallel, while the numeral split is independently sourced. About one annotated financial numeral in six does not survive translation intact. We fine-tune IndicBERT and mBERT on one language and test zero-shot on the others with content held fixed. On sustainability classification this costs 0.26–0.45 macro-F1 with IndicBERT and 0.11–0.24 with mBERT, and Hindi is the worst source language for both. mBERT outperforms IndicBERT in all twelve Hindi-source cells. Moving into a Dravidian language costs more than moving into an Indo-Aryan one, both in sustainability transfer and in translation quality. A three-stage diagnostic router assigns each failure in 72 conditions a single cause. Linguistic causes account for only 8.8–12.0% of failures, and two linguistic modules predict failure no better than chance. We therefore report the size and shape of the transfer gap and leave its causes open.

I. INTRODUCTION

Financial text in Indic languages is growing quickly, but annotated resources remain scarce [26] [32]. Multilingual pre-trained encoders make it possible to reuse labelled data from one language in another, and they transfer well on general language-understanding benchmarks [13], [14], [29]. Transfer degrades sharply, however, in low-resource settings and between distant languages [15]. Finance adds demands of its own: domain terminology, numerals written in scale conventions such as lakh and crore, and named entities that must survive a change of language.

IndicFinNLP [1] brings financial NLP to Hindi, Bengali and Telugu with three tasks: numeral detection, sustainability classification and ESG-theme classification. It was built to provide single-language baselines, not to measure transfer. When the language splits contain sentences about different topics and entities, a performance difference between languages mixes the effect of language with changes in topic, entity and difficulty. Parallel test sets exist for Indic natural language understanding [8], [9], but none covers financial or ESG text, and no multilingual ESG shared task has included an Indic language [5]. The contributions of FLUE and FLANG [30] include a financial benchmark for English and a pretrained model, but like FinBERT before, they do not happen to cover Indic languages.

This leads to our research question: do multilingual language models preserve financial meaning when knowledge transfers between Indic languages? Answering it needs two things. The first is a test set in which language is the only thing that changes. The second is an analysis of why transfer fails, in terms of the language and the financial content (terminology, numerals, morphology and tokenisation), rather than overall accuracy alone.

A second concern shapes the design. Machine-translated parallel benchmarks can silently measure translation quality instead of transfer, and low-resource languages suffer most [18], [19]. We therefore treat label validity and translation quality as quantities to measure, and we report every flagged row instead of discarding it.

This paper makes the following contributions:

1. *A controlled parallel benchmark.* We derive a four-way parallel financial corpus in Hindi, Bengali, Telugu and Malayalam from IndicFinNLP with every label intact. We audit which native splits are already parallel, translate directly between Indic languages with IndicTrans2 [21], and apply layered quality gates to all 92,760 translated rows. These gates show that about one financial numeral in six does not survive translation intact.
2. *A systematic zero-shot transfer study.* We compare three source languages (Hindi, Bengali and Telugu) and two encoders (IndicBERT [7] and mBERT [12]) across the full grid of 126 planned results per task, after validating the in-language baselines against published figures.
3. *A transfer-diagnostic framework.* We present a three-stage router. It first filters out translation noise with a semantic gate, then assigns each failure a single linguistic, data or model-behaviour cause, keeping an audit trail of every module that fired. Run over all 72

conditions, it shows that most failures are described by where they occur rather than by a linguistic property of the target text.

Section II reviews related work, Section III describes the method, Section IV reports the results and Section V concludes.

II. RELATED WORKS

We look at thirty-six peer-reviewed works grouped into four categories: NLP resources, Indic NLP resources, cross-lingual transfer and the validity of translation-based benchmarks. The numbers in brackets match the order of our literature review.

A. Financial NLP Resources and Models

IndicFinNLP [1] is the resource used in this work. It provides nine datasets with three tasks each in Hindi, Bengali and Telugu along with baseline results. Outside India, MultiFin [2] focuses on headline classification across 15 languages. It compares labeling done by speakers with a translate-then-label approach, a choice that any parallel dataset must consider. Numeracy-600K [3] highlights magnitude as a distinct skill that can be learned. It also shows that macro-F1 can differ significantly from micro-F1 when dealing with classes meaning accuracy alone may not reveal model shortcomings. FinBERT [4] demonstrates that domain- pre-training outperforms general BERT especially when training data is limited. FLUE and FLANG [30] add an English financial benchmark and a pretrained model, and the Financial PhraseBank [31] provides expert-annotated English financial sentences labelled for sentiment; neither covers Indic languages. The ML-ESG shared tasks [5] include ESG impact classification in English, Chinese, Japanese and French and they do not cover any Indic language. A recent survey of financial language models [6] points to low-resource evaluation, as ongoing challenges. A recent survey of financial language models [6] points to low-resource evaluation as an ongoing challenge**, and surveys of low-resource NLP [32] describe the approaches available when labelled data is scarce.

B. Indic NLP Resources and Benchmarks

IndicNLPSuite [7] provided IndicCorp and the first IndicBERT while IndicXTREME [8] expanded this to nine tasks across 20 languages using IndicBERT-v2. Many of its sub-tasks are n-way parallel setting an example for our corpus. Samanantar [36] provides large parallel corpora for 11 Indic languages for machine translation. IndicXNLI [9] created a parallel inference set through machine translation, following the design created by XNLI [24], and tested transfer strategies concluding that translate-train can outperform English-source transfer for Indic languages. IndoRE [10] explored transfer between Bengali, Hindi and Telugu—the same language triple as in IndicFinNLP—focusing on relation classification. Research, on part-of-speech tagging [11] measured how much target data is needed for shot Indic transfer. None of these resources include ESG text.

C. Cross Lingual Transfer

mBERT transfers across scripts but works best between languages that're similar in structure [12]. Wu and Dredze [23] likewise discovered that mBERT is able to reliably transfer zero-shot across a number of tasks and languages, although the quality of transfer varies by language, with low resource languages seemingly performing worse [33]. XLM R improved performance for languages with data by using a large scale model while showing a trade-off between good transfer and reduced capacity [13], a trade-off called the curse of multilinguality, which pre-training aims to reduce [35]. XTREME [14] measured the transfer gap as the difference between how a model works in the source language and how well it works in the target language and showed that this gap is still large; XTREME-R [29] later revised the benchmark with more challenging tasks and settings. Lauscher et al [15] found that when there is no data for the target language the transfer gets worse and this is more true for languages that are very different and that a small amount of training on the target language closes most of the gap. Choosing the source language is itself a problem that depends on how the languages are related their structure how similar their words are and how much data is available [16]. Methods that use adapters such as MAD X [17] provide a way to make the process faster and more efficient, than training. Artetxe et al. [25] have demonstrated that the transfer process can happen without a common subword lexicon or a multilingual pretraining step, while Libovický et al. [27] reported that multilingual models distinguish a language-independent component and a language-specific component. Rust et al. [28] have demonstrated that the quality of a model tokenizer for a language matters greatly.

D. Translation Based Transfer and Its Validity Risks

Machine translation forms the basis for parallel Indic benchmarks so its problems are passed along. Artetxe et al. [18] showed that translate test is more effective than earlier studies once translation quality and train test mismatch are taken into account. Agrawal et al. [19] showed that translation problems in XNLI [24] have an impact on low resource languages and that manual re annotation of Hindi and Urdu did not match the inherited English labels well. A systematic review of multilingual sentiment analysis [20] concludes that creating resource resources, from high resource ones is still a methodologically open question.

E. Positioning

We already have the methods we need. What’s missing is the right combination. Parallel Indic evaluation exists for natural language understanding [8] [9] but not for finance. Multilingual ESG work [5] does not include languages at all. Transfer gaps are usually reported in terms [14] or linked to general similarity [12] [15] [16]. Rarely connected to specific content traits, like numerals, terminology or morphology at the item level. Few-shot budgets [11] [15] and source-language selection [9] [16] have not been studied for financial text. Also there is evidence showing that unverified machine-translated benchmarks may measure translation quality of true

transfer [19] meaning any new parallel set must include clear quality checks. This work tackles all these gaps at once.

III. METHODOLOGY

The method has three parts, each one feeding into the next. First is the construction of a four-way corpus, where the translation quality is actually measured instead of being assumed. Second is the zero-shot transfer evaluation, done under a condition matrix that is fixed beforehand. Third is the attribution of each transfer failure to one cause, followed by the aggregation of all those causes. Two rules apply at every step. No row is ever taken out. If a row has similarity, empty output or detected corruption those are marked as flags, on the row. That way translation failure is always. Not silently filtered out. Also every translation, training and scoring step uses a seed.. Each step logs a hash of the configuration that produced it.

A. Source Data and Independence Audit

We begin with IndicFinNLP [1] which provides three tasks in Hindi, Bengali and Telugu: numeral-span annotation in text (task 1) binary sustainability classification (task 2) and a 10-class ESG-theme classification of news headlines (task 3). Row counts are listed in Table I. The original files are kept exactly as they are with byte-for-byte integrity preserved using SHA-256 checksums. These files are read once into Parquet format for processing.

To ensure a comparison across languages we must know whether the data splits are truly independent or if they contain the same content. We test this before any translation happens, using LaBSE [22] sentence embeddings and three methods: a structural check comparing row counts and class distributions; second cosine similarity between row-aligned pairs versus a shuffled index control to detect data that remains in order; and third, for 200 sampled source sentences we compute the maximum cosine similarity against the entire target language split to detect parallel data even after shuffling or partial row loss. Since this last measure can be hard to interpret in a domain we calibrate it using a positive control (a split confirmed to be parallel) and negative controls (data with different content but the same domain and language pair).

The audit separates the tasks into two groups. For task 1 the splits are independently sourced. The median nearest-neighbour similarity ranges from 0.58 to 0.64 which falls within the negative-control range indicating no overlap. For tasks 2 and 3 the same items appear in all three languages. In task 2 between 88.5% and 93.5% of sampled sentences have a -language match at or above $\tau = 0.82$. In task 3 a shared URL column confirms the alignment directly—532 out of 532 items match across all three languages. We use this to recover item correspondence. For task 3 we join on the URL column exactly. For task 2 we use mutual nearest-neighbour matching on LaBSE embeddings above τ. Keep only matches that are consistent across all three language pairs—1,769 items in total. Every aligned item has the gold label in all three languages. Any task-2 rows that don’t align remain in their split with an unaligned flag.

Item identifiers are based on content, than row position so a reordering of the original data cannot accidentally point an identifier to a different item. In task 1 an item is defined as a pair: (sentence span).

B. Parallel Corpus Construction

Each task is made up of three blocks, one for each language. Inside each block the native language text is translated into the three languages: Hindi, Bengali, Telugu and Malayalam. This creates nine translation directions. There are twelve splits per task and all of them use the item identifiers and gold labels (Table I). IndicFinNLP does not have any Malayalam data so Malayalam is only available, through machine translation. In each block there is one instance of Malayalam translation.

TABLE I. TRANSLATION BLOCKS AND ROW COUNTS

Block	Native source	MT targets	Task 1	Task 2	Task 3
H	Hindi	Bengali, Malayalam, Telugu	10,640	2,238	532
B	Bengali	Hindi, Malayalam, Telugu	6,130	2,228	532
T	Telugu	Hindi, Bengali, Malayalam	6,016	2,072	532
MT rows			68,358	19,614	4,788

Translation uses IndicTrans2 [21] (the indic 1B model) directly between Indic languages, with no English pivot. A pivot would add an error source and make drift impossible to attribute. One decoding configuration, beam search with five beams, a length of 256 and no sampling is applied unchanged to all nine directions so their quality figures are comparable. We chose the 1B model over its 320M distilled variant after measuring output corruption on a 180-row task-1 sample. The smaller model was eight times faster. Corrupted 10.0% of rows against 4.4%. Identical source sentences are translated once. The output shared. In task 1 where a sentence containing three numerals forms three items this reduces 68,358 translations to 40,401. Ensures each sentence has exactly one translation. The corpus holds 92,760 machine-translated rows in total.

Task 1s annotations are character offsets into the source string, which cannot survive a change of length and script. We re-locate the annotated numeral in each translation by value than by digit string so Bengali ২৪, ASCII 24 and a grouped 25,000 are all matched and record per row whether the span was recovered. A numeral rewritten as a quantity (for example 100 million, as 10 crore) is recorded separately from one that was lost.

C. Translation Quality Gates

Three automated checks are applied to every translated row. Each of these checks finds failures that the other checks do not detect.

Numeral and entity preservation. Numerals, currency symbols, percentages and lakh/crore scale terms are compared by value across digit systems. The scale word is applied on both sides

so that equivalent quantities match. A comma is treated as a thousands separator only when three digits follow it. Rows that have already been flagged as corrupted are excluded from the preservation rate. This ensures that their loss is not blamed on handling. Any direction that shows than 95% preservation is flagged.

Structural integrity. Blocking checks require every block to join four ways on the item identifier. Every translated row must carry its sources label. The native text must be free of encoding damage. Any failure in these blocking checks stops the pipeline. Non-blocking checks look for script leakage, which means characters from another script block appearing except, for the shared danda. They also detect two corruption modes that we saw in the IndicTrans2 output: entity placeholders and nukta characters that appear as literal Unicode escapes. Additionally they flag suspected truncation judged when a direction falls to half of its median length ratio. A leak is counted when the source text is clean.

Semantic similarity. Source and translation are embedded with LaBSE. Compared by cosine similarity (Eq. 1) over all 92,760 pairs with no sampling:

$$\cos(x_s, x_t) = \frac{e_s \cdot e_t}{\|e_s\| \|e_t\|} \quad (1)$$

where e's the LaBSE embedding of text x. Rows whose similarity falls below τ are flagged translation_drift. For tasks 2 and 3 the native split in the target language is a translation of the same item. This allows τ to be calibrated: a human translation is a known-good pair so τ should accept all of them. T = 0.82 accepts every pair on task 2 but rejects 23.4% on task 3 whose headlines are short and score lower for that reason alone. T is therefore set per task: 0.82 for task 2 (confirmed) 0.73 for task 3 (the value that accepts 95% of pairs) and 0.82, for task 1 inherited rather than calibrated because no human reference exists for it. Malayalam targets have no reference either.

First when a task completes all its checks the tasks twelve splits are locked into a read- release. This release contains a SHA-256 hash, for each file and a manifest that lists the translation model decoding configuration, similarity model and τ. The release datasheet is not typed by hand; instead the release datasheet is created automatically from these artefacts.

D. Evaluation Conditions

Transfer is tested on tasks 2 and 3 which're the two classification tasks. All conditions are fixed ahead of time in one condition matrix. Each task contains 21 items: nine transfer cells, one for each source–target language pair checked against native target text when available; the same nine cells checked against machine-translated target text from the source block;. Three translationese conditions that keep the language the same but change the origin (for Hindi: native translated from Bengali translated from Telugu). The nine transfer cells are grouped into four quadrants: Indo-Aryan to Indo-Aryan, Indo-Aryan to Dravidian, Dravidian to Indo-Aryan and Dravidian to Dravidian.

Because tasks 2 and 3 use the items in every language each transfer cell is scored on the same underlying content in the source and the target. Evaluation uses half of the items, chosen by a seeded hash of the item ID so every language agrees on the subset without coordination. The source model is trained on a seeded stratified fold of its native split so most evaluated items are target-language versions of sentences seen during training in the source language. This design is intentional: fixing the content on both sides of the gap lets us measure the cost of changing the language while keeping sentence difficulty constant. It is not a test of generalisation to content. A validator checks that no split is both a training source and an evaluation target within a condition. Evaluation prefers native target text so that a failure reflects transfer than translation. For Malayalam evaluation must use machine translation; the data come from a block, than the one used for training and the provenance is recorded with each result.

E. Fine-Tuning and Zero-Shot Evaluation

We compare two encoders: IndicBERT [7] a model that was pretrained on twelve Indian languages and multilingual BERT also known as mBERT (cased base) [12]. We had also planned to use XLM-R base [13] as an encoder but XLM-R base requires a fixed memory cost of 4.14 GiB for weights, gradients and optimiser state when fine-tuned with mixed-precision AdamW. This memory requirement exceeds the 3.68 GiB that our 4 GB GPU can provide at any batch size. Therefore we decided to exclude XLM-R base of altering the training protocol. Each of the encoders is used in its pretrained form. Is equipped with a single linear classification head that operates on the [CLS] representation. The models are trained using cross-entropy loss as shown in Table II. Training data consists of native human-written text. This ensures that no transfer measurement is affected by translations that we may have produced ourselves. For every configuration we run the experiment with three random seeds. The results are reported as mean ± deviation. All hyperparameters are identical for both encoders except for one detail. In the case of mBERT, on task 2 we use micro-batches of eight. Perform two-step gradient accumulation. This approach is forced by GPU memory limits. The effective batch size remains at sixteen. The gradient updates stay unchanged.

TABLE II. FINE-TUNING CONFIGURATION

Setting	**Task 2**	**Task 3**
Maximum sequence length	192	64
Batch × accumulation	16 × 1 (mBERT 8 × 2)	16 × 1
Optimiser	AdamW	AdamW
Learning rate	2×10^{-5}	3×10^{-5}
Weight decay	0.01	0.01
Warm-up	10% of steps	10% of steps
Max epochs / early-stopping patience	20 / 5	40 / 8
Dev / test fraction	0.10 / 0.10	0.15 / 0.15
Precision	fp16 mixed	fp16 mixed

Setting	Task 2	Task 3
Seeds	3	3

Each source model is first checked against IndicFinNLPs published in-language macro-F1 with a tolerance of ±0.02 and any shortfall must be diagnosed before transfer results are read. This ensures that a transfer gap cannot be an artefact of a trained source model.

Evaluation is zero-shot. Checkpoints are loaded read-only. The inference path asserts that no optimiser exists and no gradient step is taken. Every prediction is logged per instance: item, gold label predicted label and class probabilities. Macro-F1 is the metric with accuracy and per-class F1 alongside and 95 % confidence intervals come from a 1,000-sample bootstrap over items computed per seed. Following XTREME [14] the transfer gap from source s to target t is the source's in-language score minus the target score (Eq. 2),

$$\Delta_{s \to t} = \mathrm{F1}_s - \mathrm{F1}_t \tag{2}$$

computed on the underlying items wherever alignment allows so each source model is measured against its own ceiling on identical content. Because those items are largely the source model's training text the in-language score is a training-text ceiling than a held-out estimate; held-out in-language performance is reported separately as the baseline. A gap smaller than its seed standard deviation is not reported as a finding. With 21 conditions and three seeds there are 63 results per encoder and task 126 results per task. Comparisons, between source languages leave out seeds whose source model never moved from its initialisation: such source model scores badly in every language so its near-zero gap would rank it as the best source. These seeds are named in the results and remain in every other figure. Fig. 1 shows the evaluation protocol.

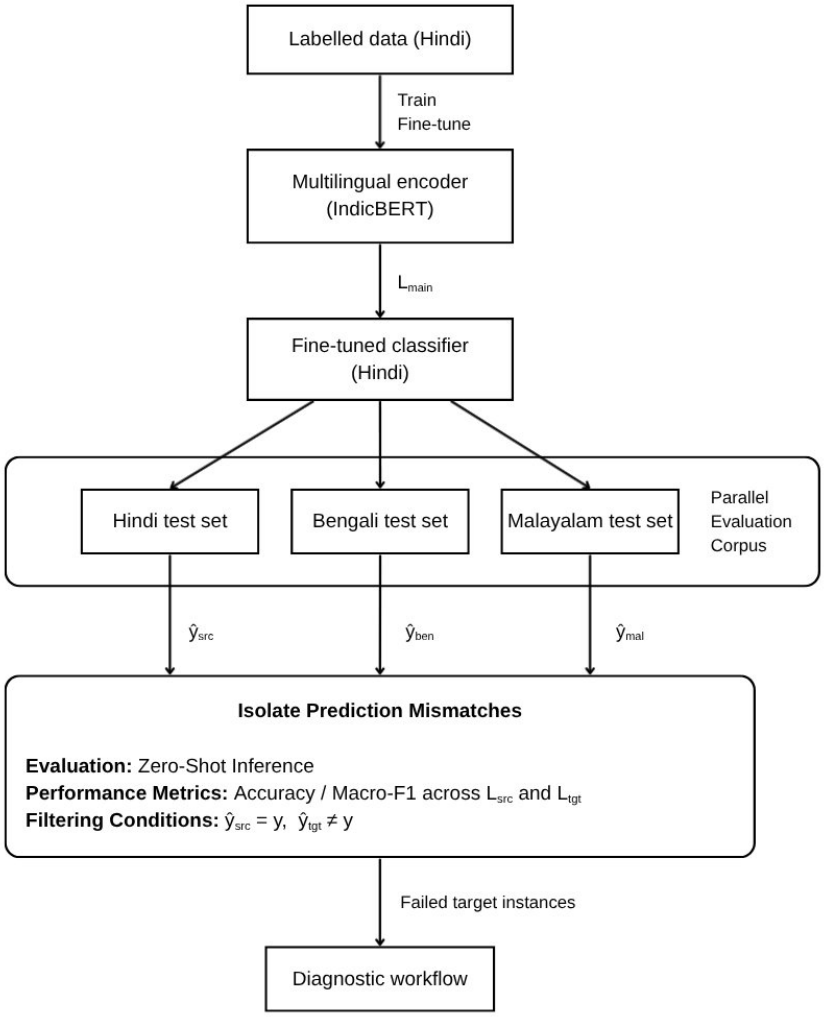


Fig. 1. Zero-shot evaluation protocol. A classifier fine-tuned on one source language (Hindi shown as the example) is run on the parallel test sets; instances correct in the source and wrong in a target pass to the diagnostic workflow.

F. Transfer-Failure Diagnostics

A transfer failure is an instance that the model classified correctly in the source language and incorrectly in the target (Eq. 3):

$$\hat{y}_s = y \ \wedge \ \hat{y}_t \neq y \tag{3}$$

Predictions are joined on (item, seed). A source prediction is never paired with a target prediction from a different seed. Since the model got this item right in the source language we can rule out lack of task ability or item difficulty by design.

Each failure gets one cause label from an ordered router along with a full audit trail showing every module that triggered. Stage 1 is a gate: if the LaBSE similarity between the source and target texts is below τ the failure is labeled translation_drift and stops there. This order matters a lot. If the linguistic modules ran first noise, from our translations might be wrongly blamed on tokenization or morphology. The gate uses the corpus similarity module as before unchanged.

Stage 2 says the remaining failures come from the way the target text is written. Quality of the tokenizer is said to influence performance in that language [28]; hence, the tokenization process is carried out using the tokenizer of the encoder (Eq. 4),

$$R_{\text{frag}} = \frac{\left|\text{tok}(x_t)\right|}{\left|\text{tok}(x_s)\right|} \tag{4}$$

and the module fires when Rfrag is 2.0 or more. Morphological masking fires when a financial word from an 18-concept ESG list (made in Hindi and translated with the corpus translation model) appears in the target inside a changed form with its root hidden by bound morphology. Stems come from the Indic NLP Librarys analyser falling back to rule-based suffix cutting when there is no analyser; Universal Dependencies does not have a Malayalam treebank. A stem matches only if it is exact with its last codepoint. With one added vowel sign. Terminology gap fires when Integrated Gradients scores, calculated against a zero-embedding baseline differ between the source item and its target counterpart. The threshold is set for each task and encoder at the percentile of divergence on pairs that the model got right in both languages and a score whose completeness error is more than 5% is marked as unconverged and never fires. A planned orthographic and numeral-mismatch module was dropped after an audit showed that items whose digit system is different between machine-translated text fail at or, below the base rate.

Stage 3 takes what Stages 1 and 2 do not explain. It points to a problem with the item or the model of the target sentence. The categories are provenance_specific, which means the item and seed works on the twin target of a different provenance. Then there is cross_lingual_hard, which means the item fails across all three targets of the source. Class_collapse happens when the run predicts one class for at least 75% of the conditions rows and the failure is that class. Seed_unstable means another seed classified the item correctly. The order of importance is translation_drift, tokenizer_fragmentation, morphological_masking, terminology_gap provenance_specific, cross_lingual_hard, class_collapse, seed_unstable and unattributed. Each label falls into one of four categories: data, model or unattributed. The unattributed part is shown as a result not hidden. The labels were checked manually on 200 failures, 50, for each target language. Fig. 2 shows the router and its ordering.

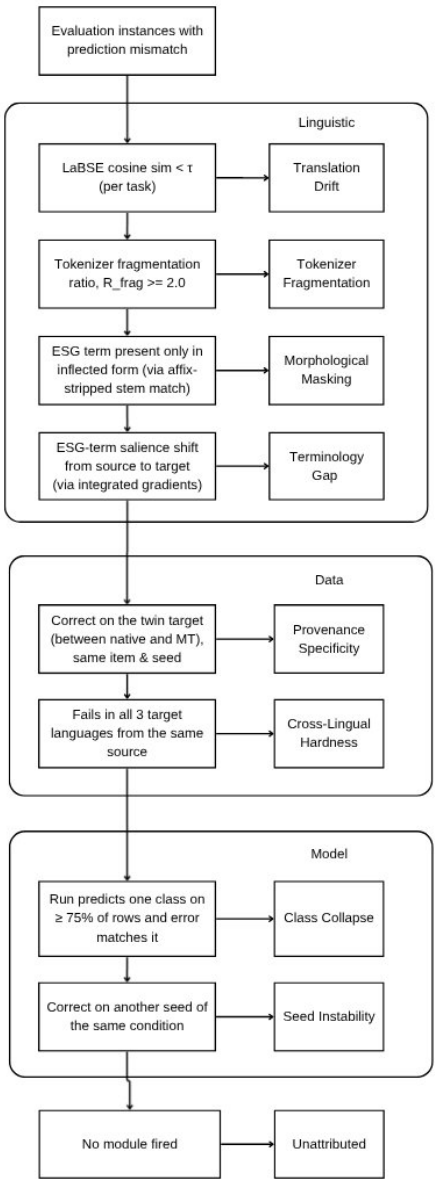


Fig. 2. Ordered diagnostic router. Each failure takes the label of the first module that fires, grouped into Linguistic, Data and Model stages; failures that no module explains stay unattributed.

G. Aggregation and Finding Rule

The unit of analysis is a failed (item, seed) row. I think a group's share of a label is simply the fraction of its failures that carry that label. Shares therefore describe what failures consist of. I notice that labels are not compared against predictions so a share does not show that a property makes failure more likely. Each share carries a 95 percent bootstrap interval that resamples (condition item) clusters so an item failing on all three seeds counts once. Groups are compared by target direction, source language, encoder and text provenance. A difference counts as a finding only if the two intervals do not overlap and the difference points the way on both encoders (on both tasks, for the encoder comparison). I added this rule before any table was read. Every

comparison is repeated without the source runs that never trained. A difference that a known measurement limit could produce on its own is reported with that caveat attached; for example Hindi targets can never receive the label because Hindi writes postpositions as separate words.

H. Reproducibility

All corpus data is stored as Parquet. Two runs of one configuration produce byte- output. Every artefact records its configuration hash. All experiments ran on a 4 GB laptop GPU (RTX 3050). That GPU decision led to three choices. First every GPU job checkpoints after each batch and resumes. Second batches are. Retried when an out-of-memory error occurs. Third jobs do not silently fall back, to the CPU. The CPU uses arithmetic, which creates translations that differ from those of the GPU, which uses float16.

IV. EXPERIMENT RESULTS

I report corpus quality first because every later number depends on corpus quality. Then I report in-language baselines, zero-shot transfer, encoder and source-language comparisons. Finally I report attribution of failures to causes. Unless stated otherwise figures shown are averages taken from three seeds, with standard deviation shown and macro-F1 is the metric.

A. Corpus Quality

All three corpora pass every blocking check. Each block joins four ways—10,640, 6,130 and 6,016 items for task 1; 2,238, 2,228 and 2,072 for task 2;. 532 Per block for task 3. No translated row has a label that differs from its source label.. No translation returned empty. Table III summarizes the -blocking findings. Task 1 is roughly an order of magnitude than the other two tasks on every measure. This makes sense given the content of task 1. It is the task that is dense with numerals, named entities and nukta characters. These are the types of elements that the two IndicTrans2 corruption modes target. Unrestored entity placeholders are the issue accounting for 5.2% of rows in task 1. Script leakage always involves a brand or domain name that was left untranslated in the original script. This is a translation failure, than damage caused by encoding issues.

TABLE III. TRANSLATION QUALITY BY TASK

Measure	**Task 1**	**Task 2**	**Task 3**
Machine-translated rows	68,358	19,614	4,788
Placeholder leakage	3,524 (5.2%)	18 (0.1%)	4 (0.1%)
Escape leakage	734 (1.1%)	1 (<0.1%)	1 (<0.1%)
Script leakage	421 (0.6%)	3 (<0.1%)	17 (0.4%)
Suspected truncation	1,141 (1.7%)	2 (<0.1%)	0
Below τ (translation_drift)	11,419 (16.7%)	357 (1.8%)	85 (1.8%)
τ	0.82 (inherited)	0.82	0.73
Median LaBSE, range over directions	0.844–0.937	0.902–0.949	0.859–0.908

Measure	Task 1	Task 2	Task 3
Entity preservation, range over directions	49.8–78.7%	89.5–95.7%	77.4–97.0%

Task 1 measures survival directly (Table IV). The annotated numeral is recovered in 82.8 percent of translations and on every row the recovered offsets slice out exactly the annotated number. Rechecking the rows with the value-level entity comparison separates the kinds of loss. In 9.7 percent of rows the output keeps digits. Not the annotated value, in 3.7 percent the digits are corrupted and in 3.3 percent there are no digits at all. 0.4 Percent are the same quantity rescaled. In thirty rows (0.04 percent) the quantity. The recovery missed it which bounds the error of the recovery procedure itself. One financial numeral, in six therefore does not survive Indic-to-Indic translation intact.

TABLE IV. SURVIVAL OF THE ANNOTATED NUMERAL (TASK 1)

Outcome	Rows	Share
Span recovered	56,624	82.8%
Digits present, annotated value gone	6,618	9.7%
Output corrupted (placeholder or escape)	2,560	3.7%
No digits in the output	2,234	3.3%
Same quantity, rescaled	292	0.4%
Quantity preserved, span missed	30	0.04%

Counting every currency symbol and percentage in a sentence not just the annotated one makes the rule stricter; rows with corruption (6.2 % of task 1) are excluded from this rate yet even so no task-1 direction reaches the 95 % preservation bar with rates ranging from 49.8 % for Bengali→Malayalam to 78.7 % for Telugu→Hindi while task-2 directions span 89.5 % to 95.7 % and task-3 directions range 77.4 % to 97.0 % and in each case the weakest direction is Hindi→Malayalam.

On LaBSE similarity task-1 divides by source language with Hindi-source directions holding a similarity of 0.906–0.937 and falling below τ on 5–11 % of rows whereas Bengali-source directions sit at 0.844–0.854 and fall below τ on 29–34 % of rows a pattern that appears only in task-1— the sole task whose blocks contain different sentences— so it reflects the source texts as much as the translation directions.

The human translations in tasks 2 and 3 make the threshold checkable as task-2's median human pair scores 0.921 with a 5th percentile of 0.846 and $\tau = 0.82$ accepting all of them whereas task-3's median is 0.870 and the 5th percentile 0.729 so $\tau = 0.82$ would reject 23.4 % of human translations a difference caused by text length (a median of 148 characters on task-2 versus 86 on task-3) because LaBSE cosine falls as context shortens and at the calibrated $\tau = 0.73$ task-3's drift-flagged rows drop from 740 to 85; a second result qualifies every similarity figure in this paper: machine

translation scores higher than human translation of the same items across all calibratable tasks and every language pair so for example on task-2 Hindi→Bengali MT scores 0.950 against 0.918 for the human Bengali–Hindi pair and LaBSE cosine rewards literal translation thus ranking directions usefully but proving to be a poor absolute measure of adequacy.

Ranking the nine directions on drift, entity preservation, corruption and truncation by the mean of per-measure ranks—so that no weights need to be invented— yields one result that holds on every task, namely translating into a Dravidian language drifts more than translating from Telugu back into an Indo-Aryan one by 2.6 points on task-1 (20.3 % against 17.6 %) 2.6 on task-2 (3.1 % against 0.5 %) and 0.6 on task-3 (2.1 % against 1.5 % at its calibrated τ); two other comparisons reverse between tasks: whether Hindi→Malayalam is worse than Bengali→Malayalam and whether Telugu-source directions drift more than, Indo-Aryan-source ones, neither of which is reported as a property of the language pair.

B. In-Language Baselines

Table V shows the in-language macro-F1 scores on each held-out test fold. Five out of the six IndicBERT baselines achieve scores at or above the figure published by IndicFinNLP. The only exception is task-2 Hindi, which's 3.7 points lower. The training behavior for this task appears training macro-F1 stays at or above 0.987 for every seed, with a test standard deviation of just 0.004. We believe the lower performance stems from our per-seed resampled test fold and unoptimized hyperparameters not from any flaw in the model itself. The comparison for task-3 adds value: the published score of 0.05 comes from a model that predicts a single class for all inputs so beating it only shows our models perform better, than random guessing.

TABLE V. IN-LANGUAGE MACRO-F1 ON HELD-OUT TEST FOLDS

Task	Language	Published [1]	IndicBERT	mBERT
2	Hindi	0.86	0.823 ± 0.004	0.874 ± 0.017
2	Bengali	0.80	0.820 ± 0.028	0.858 ± 0.014
2	Telugu	0.79	0.819 ± 0.015	0.860 ± 0.009
3	Hindi	0.05	0.153 ± 0.025	0.335 ± 0.042
3	Bengali	0.05	0.127 ± 0.055	0.292 ± 0.074
3	Telugu	0.05	0.095 ± 0.047	0.243 ± 0.036

Three of the nine task-3 IndicBERT runs never matched their training data. Bengali seed 0 and Telugu seeds 0 and 2 stayed below 0.5 training macro-F1. On a task at its data ceiling their test scores look like a run, which is why we check convergence on the training set. MBERT had no failures under identical settings. These runs stay in every all-seeds chart, for seeds that over-converged only task-3 Bengali and Telugu score 0.159 and 0.150.

C. Zero-Shot Transfer

Table VI shows the transfer gap for each source–target pair, task and encoder. There are two things to keep in mind when looking at these numbers. First the in-language score that the gap is measured against comes from the evaluation items in the source language. For task 2 80% of these items are from the source model's training fold and for task 3 it's about 70%. That means the score is not truly a test of unseen data. It's more like a training-text ceiling. For example with task-2 Hindi and IndicBERT the, in-language score is 0.954. On the held-out fold it drops to 0.823. The transfer gap shows how much performance drops compared to that training ceiling. Second task 3 has hit its data ceiling. There are 532 headlines split across ten classes. There just isn't more data to add.

TABLE VI. TRANSFER GAP (SOURCE − TARGET MACRO-F1)

Direction	**Task 2 IndicBERT**	**Task 2 mBERT**	**Task 3 IndicBERT**	**Task 3 mBERT**
Hindi→Bengali	0.448 ± 0.072	0.142 ± 0.010	0.510 ± 0.168	0.516 ± 0.049
Hindi→Telugu	0.453 ± 0.038	0.236 ± 0.056	0.514 ± 0.170	0.516 ± 0.073
Hindi→Malayalam	0.387 ± 0.057	0.209 ± 0.007	0.504 ± 0.184	0.561 ± 0.050
Bengali→Hindi	0.263 ± 0.053	0.105 ± 0.023	0.523 ± 0.163[a]	0.494 ± 0.004
Bengali→Telugu	0.404 ± 0.015	0.176 ± 0.010	0.504 ± 0.132[a]	0.523 ± 0.019
Bengali→Malayalam	0.367 ± 0.043	0.179 ± 0.028	0.515 ± 0.115[a]	0.538 ± 0.031
Telugu→Hindi	0.361 ± 0.105	0.159 ± 0.060	0.478[b]	0.489 ± 0.025
Telugu→Bengali	0.323 ± 0.047	0.196 ± 0.058	0.478[b]	0.524 ± 0.034
Telugu→Malayalam	0.417 ± 0.005	0.167 ± 0.056	0.456[b]	0.528 ± 0.054

Native target text except Malayalam, which is machine-translated from a different block. [a] Two converged seeds. [b] One converged seed, so no standard deviation.

Transfer costs take a share of in-language performance in every cell. On task 2 the differences run from 0.26 to 0.45 with IndicBERT and from 0.11 to 0.24 with mBERT. Out of Hindi IndicBERT loses 0.39 to 0.45 on task 2 and 0.50 to 0.51 on task 3. Evaluating on machine-translated target text from the sources block instead of native text moves these differences by a few points at most (Hindi source, IndicBERT task 2: 0.38 to 0.47 against 0.39 to 0.45) so the loss is not an error of our translations. On task 3 the differences cluster between 0.46 and 0.56 no matter the source, target or encoder. Fig. 3 shows the task 2 gaps for every source–target pair.

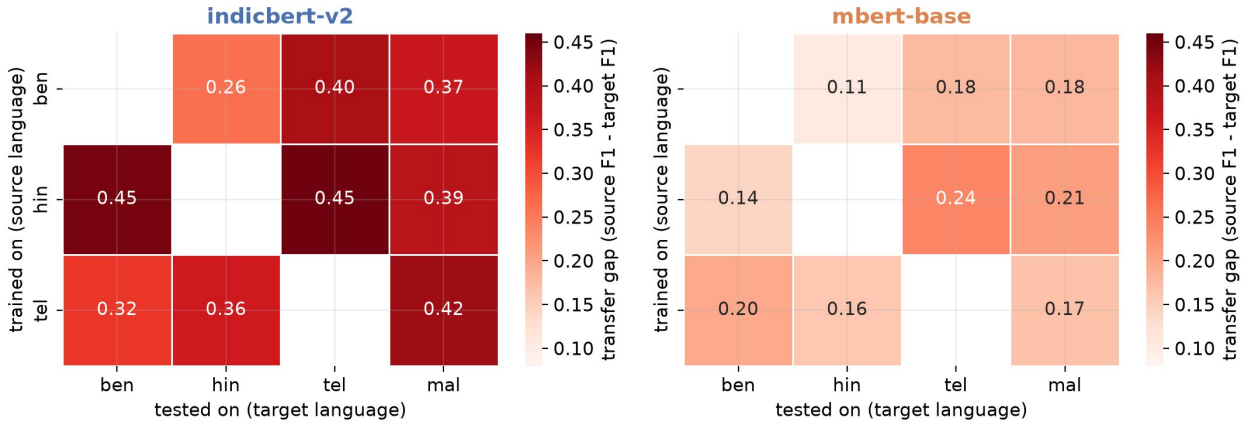


Fig. 3. Task 2 transfer gap (source minus target macro-F1) by source language (rows) and target language (columns) for IndicBERT (left) and mBERT (right).

The translationese conditions keep the language fixed and change the texts provenance. With IndicBERT on task 2 a Hindi, Bengali or Telugu model scores 5.7 to 7.8 points on machine-translated text in its own language than on native text; on task 3 the drop is 8.8 to 26.6 points. The native arm is the models split and so largely its training text while the translated arm is not. These differences are therefore a limit on the cost of translated input rather, than a direct measure of it.

D. Encoder Comparison

mBERT outperforms IndicBERT on the target language in all twelve Hindi-source cells six for each task whether the targets are native or translated. The difference in macro-F1 is between 0.15 and 0.37 which's greater than the combined seed spread of the two encoders in every cell (see Table VII). Additionally mBERT scores higher on in-language held-out data (see Table V). The two tasks show patterns. In task 2 mBERT starts at a point and loses less during transfer with gaps of 0.12 to 0.24 compared to 0.38 to 0.47 for IndicBERT. In task 3 both encoders lose the same amount, ranging from 0.45 to 0.56 for mBERT and 0.50 to 0.52, for IndicBERT. Therefore we do not claim that the better encoder always transfers better.

TABLE VII. MACRO-F1 BY ENCODER, HINDI SOURCE

	IndicBERT	**mBERT**
Task 2, in-language (held-out)	0.823 ± 0.004	0.874 ± 0.017
Task 2, target (mean of 6 cells)	0.519	0.787
Task 2, transfer gap (range)	0.38–0.47	0.12–0.24
Task 3, in-language (held-out)	0.153 ± 0.025	0.335 ± 0.042
Task 3, target (mean of 6 cells)	0.087	0.294
Task 3, transfer gap (range)	0.50–0.52	0.45–0.56

Tokenisation, a factor that influences per-language performance [28], does not explain the result [28]. IndicBERT needs 14 to 24 percent tokens than mBERT for the same sentences. Its advantage is biggest when dealing with Malayalam. Still IndicBERT performs worse on every

measure. Capacity might explain this. Model size is an essential element for multilinguality [34]. IndicBERT shares the weights of one layer across all twelve layers. So its transformer body has 7.8 million parameters. MBERT has 85.6 million parameters. To separate capacity from breadth we need to compare mBERT against XLM-R. But we could not run that comparison, on our hardware. So the reason remains unclear.

E. Source Language and Direction

Training in Hindi creates the average difference on task 2 using both encoders and training in Bengali creates the smallest difference (Table VIII). The cost is mostly in the target language. When the target is a language the average difference is 0.406 with IndicBERT and 0.194 with mBERT compared to 0.349 and 0.151 when the target is an Indo-Aryan language. Section IV-A found the direction in translation quality using a different measure. Transfer is also not equal: moving from Hindi to Bengali causes loss than moving from Bengali to Hindi on both encoders beyond the effect of random seed differences (0.448 compared to 0.263 with IndicBERT, 0.142 compared to 0.105, with mBERT).

TABLE VIII. MEAN TRANSFER GAP BY SOURCE AND TARGET FAMILY (TASK 2)

	IndicBERT	**mBERT**
Source: Bengali	0.345	0.153
Source: Telugu	0.367	0.174
Source: Hindi	0.429	0.196
Target family: Indo-Aryan	0.349	0.151
Target family: Dravidian	0.406	0.194

Typological proximity does not reliably pick the source. A same-family source is best for one of three targets with IndicBERT and for all three targets with mBERT. The two encoders also disagree about Malayalams source. The best source for Malayalam is Bengali when using IndicBERT and Telugu when using mBERT. Task 3 supports no source comparison. There is no difference, between sources and no direction asymmetry exceeds seed noise.

F. Failure Attribution

The router marked every failure in all 72 cells (two tasks, two encoders, 18 transfer conditions). Table IX shows what the failures are made of. Between 9.2% and 19.7% are not explained. Linguistic labels take up 8.8–12.0%. Data or model labels take up 69.0–79.4%. Most failures are therefore described by where they happen than by a feature of the target sentence: in every target language in only one text origin on only some seeds or under a collapsed classifier. These data and model labels show a pattern; they do not explain a process. A manual check of 200 task-2 IndicBERT labels, 50, for each target language matched 197 (98.5%). The other three (task, encoder) cells were not checked manually. Fig. 4 breaks these shares down by individual label.

TABLE IX. COMPOSITION OF TRANSFER FAILURES (SHARE OF FAILURES)

Cell	Failures	Unattributed	Linguistic	Data	Model
Task 2, IndicBERT	19,323	9.2%	12.0%	44.6%	34.2%
Task 2, mBERT	9,714	19.7%	11.3%	48.3%	20.8%
Task 3, IndicBERT	6,102	11.8%	8.8%	72.2%	7.1%
Task 3, mBERT	7,103	14.4%	9.4%	64.9%	11.2%

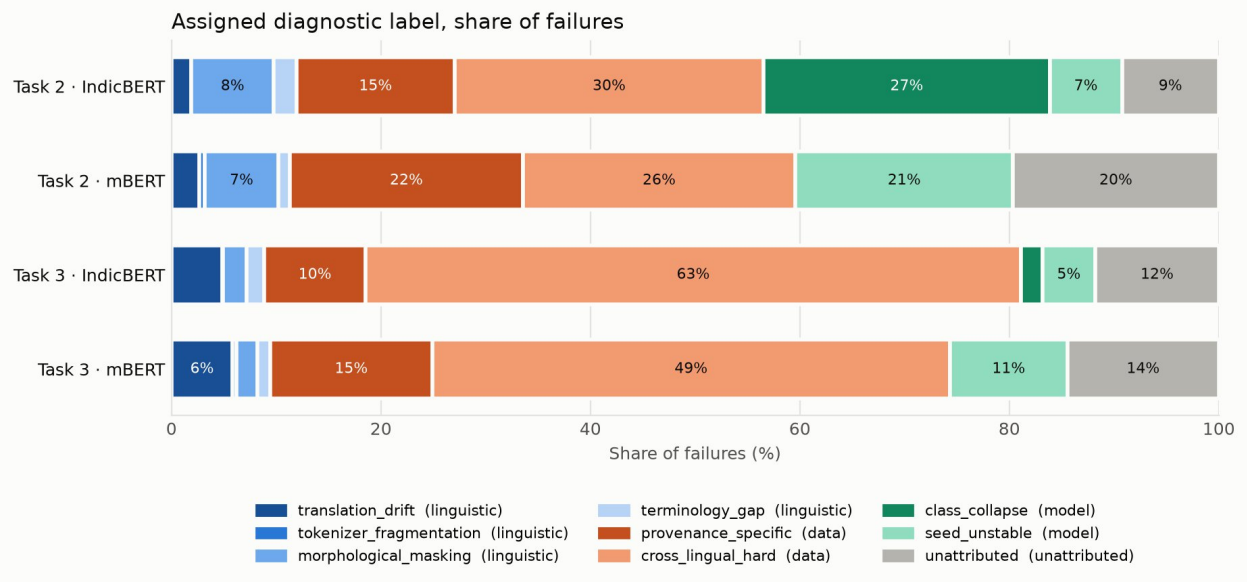


Fig. 4. Share of transfer failures assigned to each diagnostic label, by task and encoder.

Two of the modules did not validate; I noticed that these two linguistic modules did not validate. Items with a masked term do not fail often than items with the term intact and in only 20 of 44 cells do masked items fail more often. Saliency divergence does not separate failures from handled pairs; Saliency divergence shows an AUC of 0.48 to 0.52. Tokenizer fragmentation never fires, at most 0.5% of failures because Rfrag is rarely 2.0 or greater. Labels from these modules are reported, but Labels from these modules should not be read as established causes. Fig. 5 shows why: the failure and both-correct distributions largely overlap, with only 3.7–8.3% of failures above the null threshold.

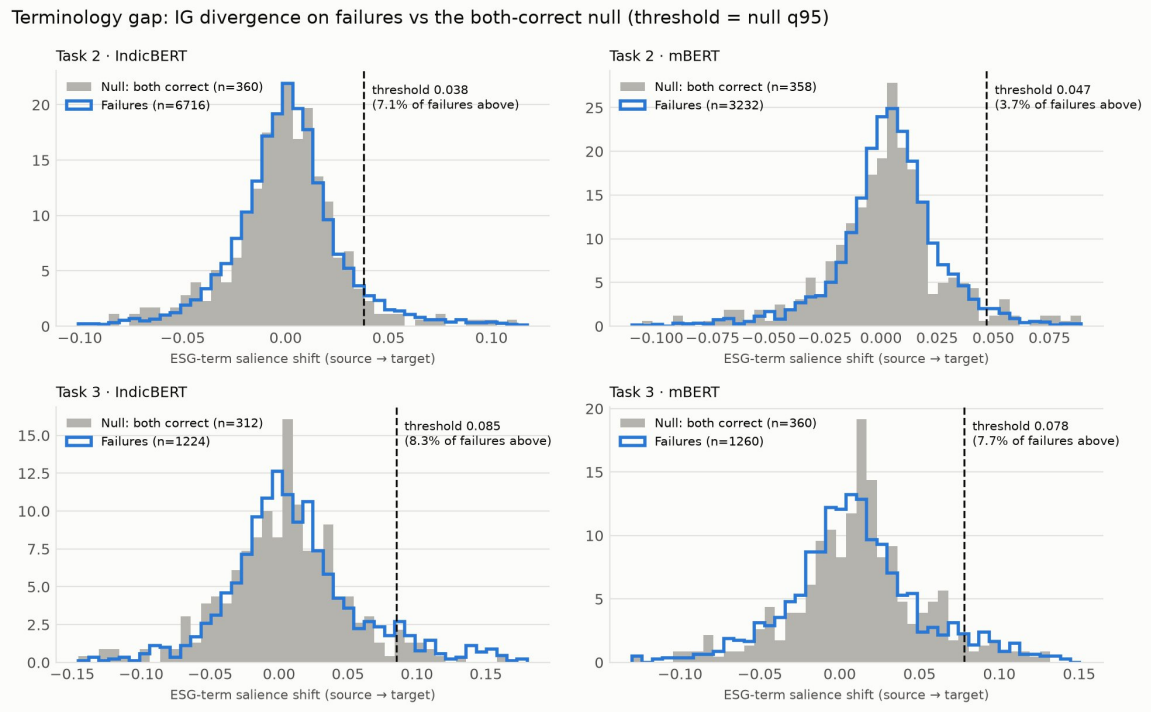


Fig. 5. Terminology gap: integrated-gradients salience shift on failures against the both-correct null. The dashed line is the null 95th percentile.

Of 228 group comparisons 22 meet the finding rule and of those 22 11 could be produced by a known measurement limit on its own. The remaining 11 fall into three groups. Direction for task 2: failures out of Telugu carry model-behaviour labels than failures into Telugu and this is true on both encoders with IndicBERT showing 40.2% against 34.4% and mBERT showing 24.9% against 14.4%; this difference mainly comes from seed instability. Source for task 2: with Bengali and Malayalam targets Telugu-source failures carry model-behaviour labels than Hindi-source failures with Telugu-source showing 46.8% against 18.4% and 22.0% against 14.7%. On task 3 Hindi-source failures carry terminology_gap labels than Telugu-source failures; however that terminology_gap label comes from a linguistic module that did not validate. Encoder: 27.3% of IndicBERT's task-2 failures are class collapse and class collapse never occurs with mBERT. IndicBERT also has more cross_lingual_hard failures on both tasks with IndicBERT showing 28.5% against 24.4% and 60.6% against 46.6%; in contrast mBERT's failures are often seed_unstable with 19.0% against 6.3% and 10.5% against 5.0% and mBERT's failures are also more often provenance_specific, with 25.5% against 15.9% and 16.6%, against 8.7%. Fig. 6 shows the group shares for each direction.

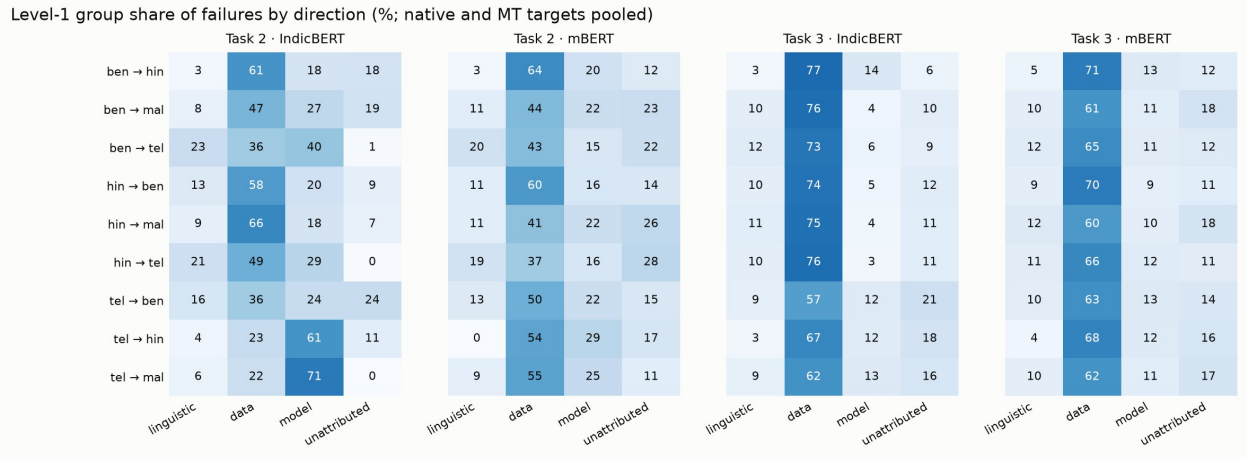


Fig. 6. Share of failures in each label group by transfer direction (%); native and MT targets pooled.

Hindi-source failures happen often than Bengali-source failures. In task two 32.9 percent of evaluated rows with IndicBERT show failures versus 27.7 percent for Bengali. With mBERT Hindi failures are 17.0 percent compared to 13.4 percent for Bengali. Yet the mix of causes does not change the rule: Hindi fails often but not for a different reason. For machine-translated targets, between nine point five and twenty-five percent of failures are marked as translation_drift or provenance_specific. That is the closest the pipeline can say that our translation caused the failure. The only differences between native and translated that satisfy the rule occur in translation_drift itself. Those are an artefact of the measurement. T accepts percent of human translations by design and LaBSE scores literal translation higher than human translation. If every comparison is repeated without the task-3 runs that never trained twenty verdicts change,. Only one changes to or, from a finding.

V. CONCLUSION AND FUTURE ENHANCEMENT

A. Conclusion

We wondered whether multilingual language models keep meaning intact when knowledge moves between Indic languages. To answer this question we built a four-way financial corpus in Hindi, Bengali, Telugu and Malayalam using IndicFinNLP. The corpus contains ninety-two thousand seven hundred sixty machine-translated rows. Each row is. Flagged instead of being filtered. Every row is checked for preservation, structural integrity and semantic similarity using a threshold that was set based on human translations. The independence audit that came before this work changed the design. The sustainability and ESG-theme splits turned out to be the items, in every language. The numeral split however is independently sourced.

Financial numerals are fragile when translated. One in six annotated numerals does not stay the same during Indic-to-Indic translation. No translation direction keeps 95% of the numerals currency symbols or percentages in a sentence. Transferring across languages comes with a cost. Even when the content stays the same IndicBERT loses 0.26 to 0.45 in macro-F1 for sustainability

classification and mBERT loses 0.11 to 0.24. Using our translations instead of native text changes the gap only slightly so the drop is not due to the dataset. MBERT performs than IndicBERT in all twelve Hindi-source cases. The difference does not come from tokenization. For sustainability classification Hindi is the source language for both models. Here the cost mainly comes from the target language. Switching to a language costs more than switching to an Indo-Aryan one. The same pattern shows up in translation quality across all three tasks, measured by a method. The ESG-theme task, with 532 headlines, across ten classes is too small to compare sources properly.

The diagnostic framework labeled every failure in all 72 (task, encoder, condition) cells. The labels point out what went wrong. They don't explain why. Linguistic labels cover 8.8 to 12.0 percent of the failures. Most of the failures are described by where they happen. In every target language on only some seeds or under a collapsed classifier. Two linguistic modules, masking and terminology gap did not predict failure any better than guesswork. Because of that we report the size and shape of the transfer gap, with confidence.. We leave the causes of the transfer gap as open questions.

Several limits bound these conclusions. Malayalam has no text so Malayalam results combine transfer difficulty with translation quality. Human verification covers a sample of 180 translated rows per task, not the corpus. LaBSE similarity ranks directions reliably. Labse rewards literal translation so LaBSE is not an absolute quality score. Training and evaluation share content across languages by design. The gaps measure the cost of changing language and not generalisation to unseen sentences.. A single 4 GB GPU ruled out XLM-R the comparison that would separate model capacity, from multilingual breadth.

B. Future Enhancement

Encoders. Running XLM-R [13] against mBERT on hardware would let us see the effect of multilingual breadth while keeping capacity the same and would decide if mBERTs better performance comes from more capacity. The BERT-based IndicBERT-v2 [8] would ask the question, for an Indic-specific model that has a full-size transformer body.

Human verification. Native-speaker review of each target language beyond the maintainer sample would show how well the work is done. Native-speaker review can reveal more than automated checks, which only confirm that numerals, labels and script survived. A small native Malayalam test set would separate Malayalam transfer, from Malayalam translation.

Diagnostics. Comparing each labels failure rate with its prediction rate would reveal whether a property makes failure more likely something that looking at failures alone cannot show. The morphology module and the terminology module need either a redesign or an intervention test, such, as restoring a term and then measuring how the prediction changes. The numeral module, which was left out of this build could be rebuilt using the span annotations from task 1.

Evaluation. Task 1 is the one that comes from a source. If we treat detection as a transfer task we can get a measurement that does not include shared content. Training on one part of the data and evaluating on the part would give a held-out estimate in addition, to the fixed-content gap that is reported here. Few-shot target-language training [11] [15]. Translate-train [9] [18] would show how much of the gap a small amount of target data or translated training data can recover. For the ESG-theme task, paraphrase augmentation, the method used by IndicFinNLP [1] would lift the task above its data ceiling.

Coverage. IndicTrans2 covers all 22 scheduled languages [21] so the pipeline can be extended to other Dravidian and Indo-Aryan languages by simply changing configuration. This will test whether the, into-Dravidian penalty remains true beyond Telugu and Malayalam.

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